CECH-TO-COHOMOLOGY SPECTRAL SEQUENCE FOR A SURJECTION OF PRESHEAVES OF SETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction, including the resolution property and the identification of the E_1 terms.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Sheafification of a presheaf of sets on a site and representable presheaves. Same chapter, the sheafification of presheaves of sets and the resulting surjectivity on sheaves. Supplies the hypothesis on the map K' -> K and the construction of the fibre products K'_p.
- Spectral sequence of a simplicial object or of a double complex. Homology, the spectral sequence attached to a simplicial abelian group or to the double complex of the augmented Cech construction. Supplies the spectral sequence whose E_1 and E_2 identifications give the statement.

Source reference. Original lemma 03CI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2065–2065. Navigation pointer invoked at proof line 2065; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1475–1475. Navigation pointer invoked at proof line 2092; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $K' \to K$ be a map of presheaves of sets on $\mathcal{C}$ whose sheafification is surjective. Set $K'_p = K' \times_K \ldots \times_K K'$ ($p+1$-factors). For every abelian sheaf $\mathcal{F}$ there is a spectral sequence with $E_1^{p,q} = H^q(K'_p, \mathcal{F})$ converging to $H^{p+q}(K, \mathcal{F})$.*

Proof. Since sheafification is exact, we see that $(K'_p)^\#$ is equal to $(K')^\# \times_{K^\#} \ldots \times_{K^\#} (K')^\#$ ($p+1$-factors). Thus we may replace K and K' by their sheafifications and assume $K \to K'$ is a surjective map of sheaves. After replacing $\mathcal{C}$ by a "larger" site as in Sites, Lemma 03CI we may assume that K, K' are objects of $\mathcal{C}$ and that $\mathcal{U} = \{K' \to K\}$ is a covering. Then we have the Čech to cohomology spectral sequence of Lemma 03AZ whose E_1 page is as indicated in the statement of the lemma. □